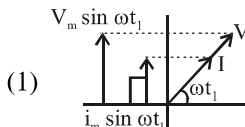
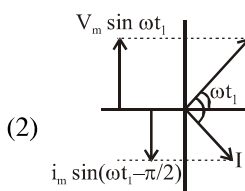


ALTERNATING CURRENT

1. r.m.s. value of A.C. is defined in terms of
 - (1) heating effect of current
 - (2) chemical effect of current
 - (3) magnetic effect of current
 - (4) none of these
2. An applied voltage signal consists of a superposition of a d.c. voltage and an a.c. voltage of high frequency. The circuit consists of an inductor and a capacitor in series, then :
 - (1) d.c. signal will appear across L and C
 - (2) d.c. signal will appear across C and a.c. signal across L.
 - (3) d.c. signal will appear across L and a.c. signal across C
 - (4) across L and C a.c. signal will appears
3. Which is true for a.c.
 - (1) r.m.s value is non-zero
 - (2) average values is zero
 - (3) instantaneous values may be zero
 - (4) All of above
4. Circuits used for transporting electric power, requires :
 - (1) low power factor
 - (2) high power factor
 - (3) zero power factor
 - (4) any values of power factor
5. More selective a.c. circuit should have :
 - (1) low R, high L
 - (2) low R, low L
 - (3) high R, high L
 - (4) high R, low L
6. Metal detector works on principle of
 - (1) Newton's third law
 - (2) Resonance in a.c. circuit
 - (3) heating effect of current
 - (4) electrostatic shielding effect
7. Which is true for series R-L-C circuit :
 - (1) The power rating of an element used in a.c. circuit refers to its average power rating.
 - (2) The power consumed in an a.c. circuit is never negative
 - (3) The power factor in a R-L-C circuit is a measure of how close the circuit is to expanding the maximum power.
 - (4) All
8. A choke coil is used for limiting current in
 - (1) dc circuit only
 - (2) ac circuit only
 - (3) in both ac and dc
 - (4) electronic valves
9. Ohm's law expressed as $V = IR$
 - (1) may never be applied to ac
 - (2) applied to ac in the same manner as to dc.
 - (3) always applies to ac circuits when z is substituted for R
 - (4) tell us that $V_{\text{eff}} = 0.707 V_{\text{max}}$ for ac
10. To express AC power in the same from as DC power, a special value of current is defined and used, is called
 - (1) root mean square current (I_{rms})
 - (2) effective current
 - (3) induced current
 - (4) both (1) and (2)
11. Which of the following figure shows that the current phasor I is $\frac{\pi}{2}$ behind the voltage phasor V .



(1)



(2)

 - (3) Both (1) and (2)
 - (4) Neither (1) nor (2)

12. Equation $\frac{di}{dt} = \frac{V}{L} = \frac{V_m}{L} \sin \omega t$ implies that the equation for $i(t)$, the current as a function of time, must be such that

(1) its slope $\frac{di}{dt}$ is a sinusoidally varying quantity with the same phase as the source voltage

(2) an amplitude given by $\frac{V_m}{\omega L}$

(3) Both (1) and (2)

(4) Neither (1) nor (2)

13. The phenomenon of resonance is common among system that have a tendency

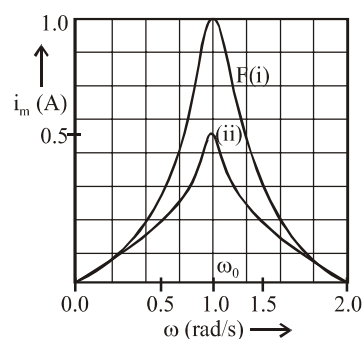
(1) to oscillate at a particular frequency

(2) to get maximum amplitude

(3) Both (1) and (2)

(4) Neither (1) and (2)

14. Figure shows the variation of i_m with ω in a



- (1) R–L–C circuit (2) R–L circuit
(3) R–C circuit (4) None of these

15. In tuning we vary the capacitance of a capacitor in the tuning circuit such that the resonant frequency of the circuit becomes nearly equal to the frequency of the radio signal received. When this happens, theA..... with the frequency of the signal of the particular radio station in the circuit is maximum. Here, A refers to

- (1) Resonant frequency
(2) Impedance
(3) Amplitude of the current
(4) Reactance

16. An alternating emf of frequency $\nu = \left(\frac{1}{2\pi\sqrt{LC}} \right)$

is applied to a series L–C–R circuit. For the frequency of the applied emf

(1) the circuit is at resonance and its impedance is made up only of a reactive part

(2) The current in the circuit is in phase with the applied emf and voltage across R equals the applied emf.

(3) the sum of the potential difference across the inductance and capacitance equals the applied emf which is 180° ahead of phase of the current in the circuit

(4) the quality factor of the circuit is $\frac{\omega L}{R}$ or

$\frac{1}{\omega CR}$ and this is measure of the voltage magnification (produced by the circuit) at resonance as well as the sharpness of resonance of the circuit

17. A value of ω for which the current amplitude is

$\frac{1}{\sqrt{2}}$ times its maximum value. At this value the

power dissipated by the circuit becomes

- (1) double (2) one-fourth
(3) one-third (4) half

18. When a capacitor (initially charged) is connected to an inductor, the charge on the capacitor and the current in the circuit exhibit the phenomenon of

- (1) electrical oscillations
(2) induction
(3) power factor
(4) All of these

19. Choose the incorrect

- (1) Phasor diagram say nothing about the initial condition
(2) Any arbitrary value of t , draw different phasor which show the relative angle between different phasors. The solution, so obtained is called the steady-state solution
(3) We do have a transient solution which exists even for $\nu = 0$. The general solution is the sum of transient solution and the steady-state solution
(4) None of the above

20. Choose the **INCORRECT** options

- (1) If the resonance is less sharp, not only is the maximum current less, the circuit is close to resonance for a larger range $\Delta\omega$ of frequencies and the tuning of the circuit will not be good.
- (2) Less sharp the resonance less is the selectivity of the circuit or vice-versa
- (3) If quality factor is large i.e. R is low or L is large the circuit is more selective
- (4) None of the above

21. L–C oscillation is not realistic for the following reasons. Which of following reasons is **INCORRECT**?

- (1) Every inductor has some resistance
- (2) The effect of resistance is to introduce a damping effect on the charge and current in the circuit and the oscillations finally dies away.
- (3) Even if the resistance is zero, the total energy of the system would not remain constant. It is radiated away from the system in the form of electromagnetic waves
- (4) None of the above

22. Same current is flowing in two alternating circuits. The first circuit contains only inductance and the other circuit contains only a capacitance. If the frequency of the e.m.f. of A.C. is increased, the effect on the value of the current will be

- (1) increase in the first circuit and decrease in the other
- (2) increased in both circuits
- (3) decrease in both circuits
- (4) decrease in the first circuit and increase in the other

23. Non-resonant circuit, what will be the nature of circuit for frequencies higher than the resonant frequency ?

- (1) Resistive
- (2) Capacitive
- (3) Inductive
- (4) None of these

24. A choke is preferred over a resistance for limiting current in AC circuit because

- (1) choke is cheap
- (2) there is no wastage of power
- (3) choke is compact in size
- (4) choke is a good absorber of heat